Fluid Phase Equilibria, 51 (1989) 353-367
Elsevier Science Publishers B.V., Amsterdam — Printed in The Netherlands

THE VOLUMETRIC ANALYSIS AND PREDICTION OF LIQUID-LIQUID-VAPOR EQUILIBRIA IN CERTAIN CARBON DIOXIDE + n-ALKANE SYSTEMS

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ABSTRACT

This article deals with the use of a volumetric technique for obtaining compositional and density data of the L_1L_2V equilibria of the systems CO_2 + tetradecane, CO_2 + pentadecane and CO_2 + hexadecane. In these systems a barotropic inversion occurs between the two liquid phases. Primary attention has been given to the influence of experimental inaccuracies on the results. The phase behaviour of the systems CO_2 + tetradecane, CO_2 + pentadecane and CO_2 + hexadecane has been fitted to the Peng-Robinson equation of state and a modified Peng-Robinson equation of state showing no significant difference between these equations of state.

INTRODUCTION

Carbon dioxide + n-alkane systems are, for example, of interest to enhanced oil recovery (Gardner et al., 1981) and supercritical extraction (McHugh and Krukonis, 1986). Much research has been done on the high pressure phase behaviour of CO_2 + n-alkane systems. The phase behaviour of these systems can be divided into the following five groups:

1: CO_2 + methane up to CO_2 + hexane. In these systems no liquid-liquid immiscibility occurs. The system CO_2 + ethane exhibits azeotropy (Poetmann and Katz, 1945). The critical curve (L=G) of these systems is a continuous curve between the critical points of the pure components.

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2: CO_2 + heptane up to CO_2 + dodecane. CO_2 + heptane is the first member of the CO_2 + n-alkane series to exhibit limited miscibility in the liquid phase (Im and Kurata, 1971). The systems of this group exhibit a L_1L_2V locus starting from a quadruple point SL_1L_2V and ending in an upper critical end point (UCEP) (L_1=L_2+V).

3: CO_2 + tridecane. This system exhibits two L_1L_2V loci as mentioned by Enick et al. (1985). One three phase curve starts from the quadruple point (SL_1L_2V) and ends in an UCEP (L_1=L_2+V). The other three phase locus starts from a lower critical end point (LCEP) (L_1=L_2+V) and ends in an UCEP (L_2+L_1=V).

4: CO_2 + tetradecane up to CO_2 + heneicosane (Hottovy et al., 1981). The L_1L_2V loci exhibited by the systems of this group all start from a quadruple point SL_1L_2V and end in an UCEP (L_2+L_1=V). CO_2 + heneicosane is the last member of the CO_2 + n-alkane series to exhibit L_1L_2V immiscibility.

5: CO_2 + C_{22}^+ (Fall and Luks, 1984). In this group no liquid-liquid immiscibility occurs.

The present study deals with the measurement of L_1L_2V equilibria of the systems CO_2 + tetradecane, CO_2 + pentadecane and CO_2 + hexadecane by using the volumetric method as proposed by Knobler and Scott (1980). The major aim of this study is to investigate the accuracy of the volumetric technique and to obtain three phase compositional data of the systems CO_2 + tetradecane, CO_2 + pentadecane and CO_2 + hexadecane. For the system CO_2 + hexadecane, to our knowledge, no compositional data are available.

The volumetric method as proposed by Knobler and Scott is applicable to systems of which all chemical potentials are fixed and the volume of every phase can be measured. Because of this the method is not applicable to the measurement of phase equilibria in which a solid phase is present. In this case the method should be used with trace amount of solid phase being present (Yang et al., 1976). By fixing the temperature and/or pressure the compositions of the coexisting phases can be calculated by solving the mass balance equations for several experiments using the well-known least squares method. Because of the inevitable variance in the fixed temperatures and/or pressures the composition of a phase slightly differs between the experiments. This variance $\Delta x_{i,j}$ in the mole fraction $x_{i,j}$ of component i in phase j follows from eqn. (1):

$$\Delta x_{i,j}=(\delta x_{i,j}/\delta T)\Delta T+(\delta x_{i,j}/\delta p)\Delta p \qquad (1)$$

Because in a phase the sum of the mole fractions equals unity the following equation must hold:

$$\sum_{i=1}^{n} \Delta x_{i,j}[(\delta x_{i,j}/\delta T)\Delta T+(\delta x_{i,j}/\delta p)\Delta p]=0 \qquad (2)$$

From this equation it can be concluded that the volumetric method can fail if it is used near a critical point in a multi-component system. According to our experience corrections for the shape of the fluid interfaces have also to be taken into account. This is done by assuming the shape of the fluid interfaces to be elliptic. The influence of the accuracy of the parameters used by this method on the final results is calculated by performing a parameter sensitivity analysis on all parameters (measured volumes, masses of the components, calibration constants of the equilibrium cell, the model used for the fluid interfaces). The experimental results obtained for CO_2 + tetradecane and CO_2 + pentadecane can be compared with the results obtained by Hottovy et al.(1981).

The obtained compositional data have been fitted to the well-known Peng-Robinson equation of state (Peng and Robinson, 1976) and a modified Peng-Robinson equation of state (Tsonopoulos and Heidman,1986) using the classical van der Waals mixing rules.

EXPERIMENTAL

All experiments were carried out in an apparatus for use with pressures up to 20 MPa and temperatures up to 473 K described earlier by de Loos et al. (1983). With this so called Cailletet apparatus which is shown in Figure 1, volumetric measurements can be carried out by visual determination of the height of the fluid interfaces relative to the top of the cylindrical equilibrium cell (C) (Figure 2). The dimensions of the Cailletet-tube used are the same as the dimensions of the equilibruim cell used by Specovius et al. (1981). The internal diameter is about 1 cm and the volume available for the mixture is 8 to 16 cm^3. The sample can be agitated by small stirrers which consist of a soft iron rod, sealed in with teflon with a measured volume of ± 0.015 cm^3 each. These magnets are activated by two button magnets B which move up and down. The glass capillary tube is kept at the desired temperature by a thermostat with circulating water. The temperature was measured to an accuracy of 0.01 K using a platinum resistance thermometer. During each experiment the temperature was kept constant to better than 0.03 K. Mercury is used as an inert pressure intermediate between the sample and the hydraulic oil in the pressure generating system.The pressure is measured with a Heise gauge, calibrated against a dead-weight gauge, with an accuracy of 0.01 MPa. During an experiment the mixture is brought to the desired temperature and stirring was stopped. After phase separation was completed the heights of the interfaces were measured relative to the top of the tube C by a cathetometer with an accuracy of 0.05 mm. To be able to account for the shape of

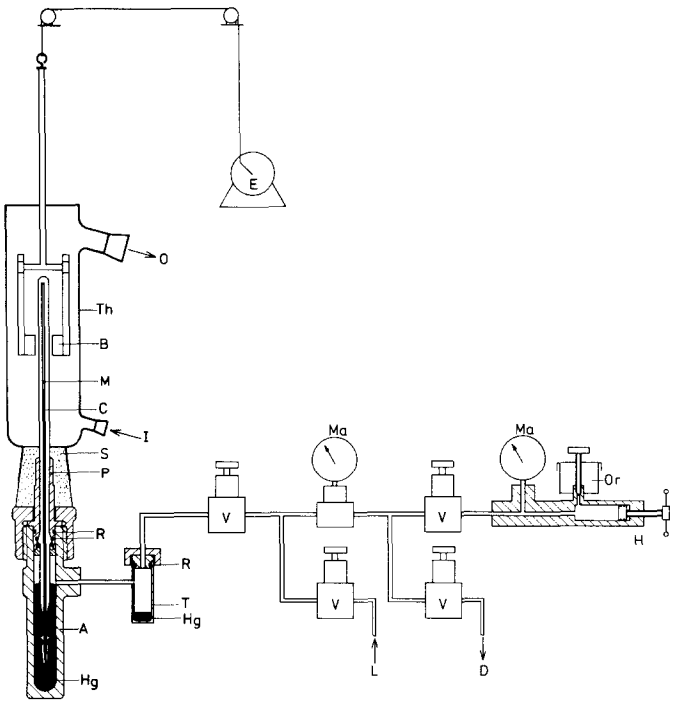


Fig. 1. The Cailletet equipment: A, autoclave; B, button magnets; C, Cailletet tube; D, drain; H, rotating handpump; Hg, mercury; I, thermostat liquid in; L, line to dead weight pressure gauge; M, mixture being investigated; Ma, manometers; O, thermostat liquid out; Or, hydraulic oil reservoir; P, closing plug; R, Viton O-rings; S, silicone rubber stopper; T, mercury trap; Th, glass thermostat; V, valve.

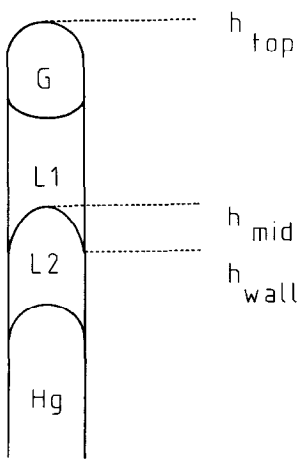


Fig. 2. Detail of the equilibrium cell: h_{top}, height of the top of the cell (used as reference); h_{mid}, height of the interface in the middle of the equilibrium cell; h_{wall}, height of the interface at the wall of the equilibrium cell; $\Delta h_m = h_{mid} - h_{wall}$; d, internal diameter of the equilibrium cell.

the fluid interfaces the height of every interface is measured in the middle and at the wall of the Cailletet tube. The tube is used in an upright position so the section perpendicular to the axis of the tube is circular. The section through the axis of the tube is supposed to be an ellipse and the contact angle between the fluid interface and the glass wall is supposed to be zero. The formula used to account for the volume of a fluid interface (half the volume of an ellipsoid) is (see Fig. 2):

$$V_m = 2*\Pi*d^2*\Delta h_m/3 \qquad (3)$$

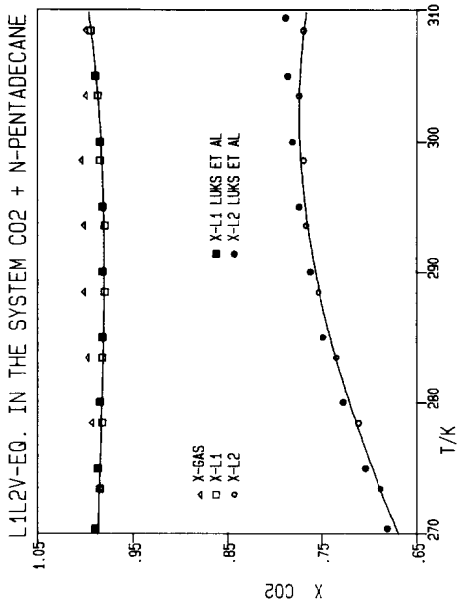
The Cailletet tube has been calibrated to an estimated accuracy of 0.01 ml. After determination of the phase volumes the total volume of the sample in the tube was varied by changing the amount of mercury in the tube. After reaching equilibrium the procedure was repeated. This procedure was used once at every temperature and with every filling of the tube.

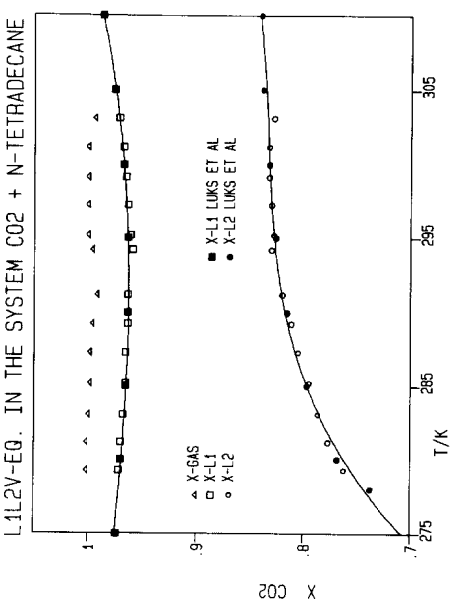
The mixture in the tube can be prepared with an accuracy of 0.01 gr of every component. The materials used are: carbon dioxide (99.99%) supplied by Matheson; tetradecane and hexadecane obtained from Fluka with a stated purity of respectively 99% and 98%; pentadecane obtained from Merck with a stated purity of 99%. All materials were used without further purification.

RESULTS

The mole fractions CO_2 and densities obtained by using the volumetric method are listed in Tables 1 to 3 and plotted in Figures 3 to 5. The calculated 95%-confidence intervals are listed in Tables 4 to 6. Another way of estimating confidence intervals is the method used by Hottovy et al. (1981): the calculation of the averaged absolute deviations (AAD). The AAD are also listed in Tables 4 to 6. From Tables 4 to 6 it can be concluded that the accuracy in the calculated mole fractions is approximately ±0.003 for the vapor phase, ±0.002 for the L_1 phase and ±0.003 for the L_2 phase. The accuracy in the calculated densities is approximately ±0.008 gml^{-1} for the vapor phase, ±0.005 gml^{-1} for the L_1 phase and ±0.007 gml^{-1} for the L_2 phase. From Fig. 3 it can be concluded that, at higher temperatures, there is a small but significant difference between our data and those of Hottovy et al. (1981). This difference is probably due to the fact that these authors, who in principle used the same experimental technique, assumed the vapor phase to be pure carbon dioxide. However near the UCEP this assumption is not correct.

A performed parameter sensitivity analysis has shown that all parameters were obtained accurately enough, because the sensitivity of the calculated





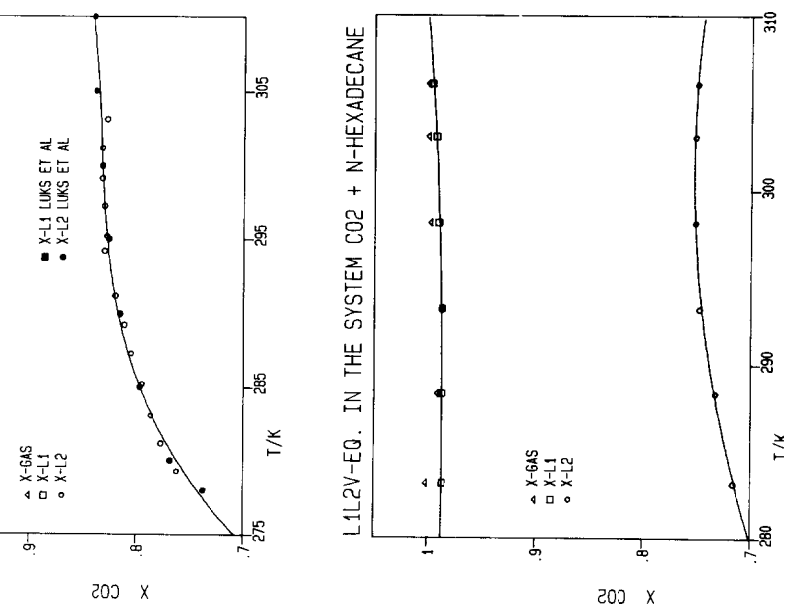


Fig. 3. Experimental compositions (see text) of the systems $[(1-x)\ C_nH_{2n+2} + x\ CO_2]$ for n=14, 15, 16 at three phase equilibria L_1L_2V as a function of temperature. Δ, □, ○: this work; □, ○: Hottovy et al.(1981). a: CO_2 + tetradecane; b: CO_2 + pentadecane; c: CO_2 + hexadecane.

TABLE 1

L_1L_2V equilibria in the system [(1-x) $C_{14}H_{30}$ + x CO_2]. Experimental pressure p, mole fraction x and density ρ as a function of temperature T.

T/K	p/MPa	x_V	x_{L_2}	x_{L_1}	ρ_V/gml^{-1}	ρ_{L_2}/gml^{-1}	ρ_{L_1}/gml^{-1}
279.26	3.980	1.0022	0.7622	0.9726	0.1204	0.8495	0.8997
281.14	4.179	1.0023	0.7770	0.9693	0.1298	0.8517	0.8848
283.08	4.391	1.0003	0.7860	0.9675	0.1344	0.8494	0.8779
285.16	4.607	0.9993	0.7947	0.9663	0.1455	0.8466	0.8712
287.24	4.829	0.9985	0.8040	0.9648	0.1548	0.8461	0.8595
289.21	5.052	0.9961	0.8118	0.9641	0.1548	0.8460	0.8554
291.18	5.287	0.9929	0.8189	0.9631	0.1658	0.8460	0.8427
294.21	5.658	0.9962	0.8298	0.9595	0.1733	0.8396	0.8380
295.17	5.781	1.0014	0.8274	0.9618	0.1853	0.8334	0.8402
297.26	6.052	1.0013	0.8305	0.9630	0.1998	0.8310	0.8276
299.19	6.325	1.0013	0.8319	0.9652	0.2149	0.8308	0.8089
301.15	6.615	1.0015	0.8316	0.9679	0.2334	0.8267	0.7972
303.13	6.919	0.9945	0.8281	0.9708	0.2824	0.8097	0.7887

TABLE 2

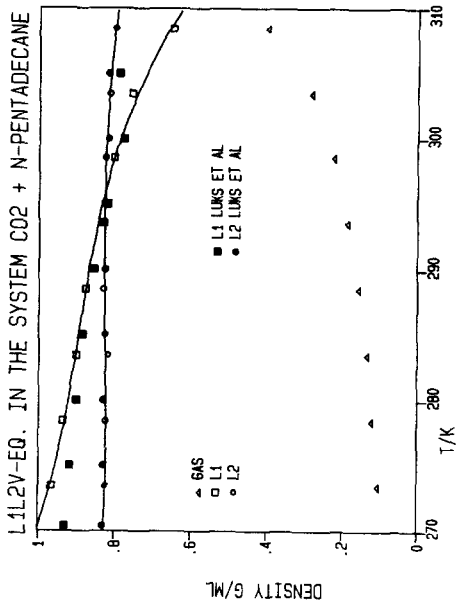
L_1L_2V equilibria in the system [(1-x) $C_{15}H_{32}$ + x CO_2]. Experimental pressure p, mole fraction x and density ρ as a function of temperature T.

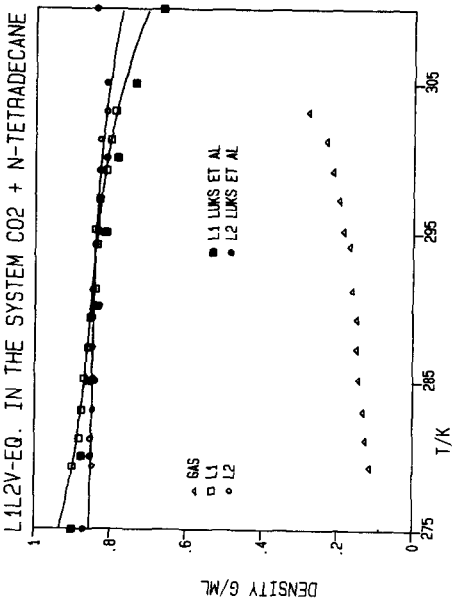
T/K	p/MPa	x_V	x_{L_2}	x_{L_1}	ρ_V/gml^{-1}	ρ_{L_2}/gml^{-1}	ρ_{L_1}/gml^{-1}
273.42	3.453	0.9843	0.6888	0.98465	0.1085	0.8234	0.9647
278.48	3.911	0.9935	0.7123	0.9825	0.1212	0.8233	0.9374
283.47	4.435	0.9988	0.7353	0.9817	0.1366	0.8318	0.9019
288.49	4.995	1.0026	0.7535	0.9793	0.1584	0.8276	0.8777
293.54	5.602	1.0034	0.7674	0.9785	0.1856	0.8325	0.8292
298.55	6.275	1.0043	0.7707	0.9832	0.2226	0.8213	0.8022
303.48	7.019	1.0012	0.7741	0.9864	0.2838	0.8109	0.7517
308.43	7.842	0.9997	0.7699	0.9927	0.4000	0.8015	0.6489

TABLE 3

L_1L_2V equilibria in the system [(1-x) $C_{16}H_{34}$ + x CO_2]. Experimental pressure p, mole fraction x and density ρ as a function of temperature T.

T/K	p/MPa	x_V	x_{L_2}	x_{L_1}	ρ_V/gml^{-1}	ρ_{L_2}/gml^{-1}	ρ_{L_1}/gml^{-1}
283.17	4.440	1.0022	0.7157	0.9868	0.1195	0.8351	0.9081
288.33	5.002	0.9931	0.7326	0.9870	0.1545	0.8372	0.8665
293.18	5.624	0.9888	0.7474	0.9869	0.1763	0.8311	0.8312
298.13	6.304	0.9987	0.7506	0.9898	0.2154	0.8305	0.7826
303.11	7.056	1.0008	0.7523	0.9923	0.2826	0.8209	0.7236
306.10	7.548	1.0008	0.7500	0.9955	0.3501	0.8169	0.6561





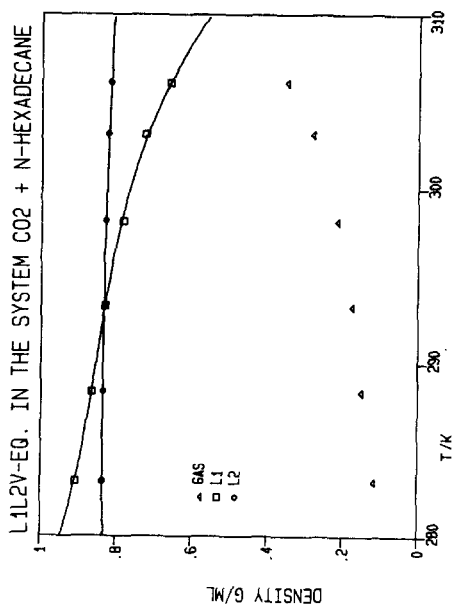


Fig. 4. Experimental densities (see text) of the systems [(1-x) C_nH_{2n+2} + x CO_2] for n=14, 15, 16 at three phase equilibria L_1L_2V as a function of temperature. Δ, □, O: this work; □, O: Hottovy et al.(1981). a: CO_2 + tetradecane; b: CO_2 + pentadecane; c: CO_2 + hexadecane.

TABLE 4

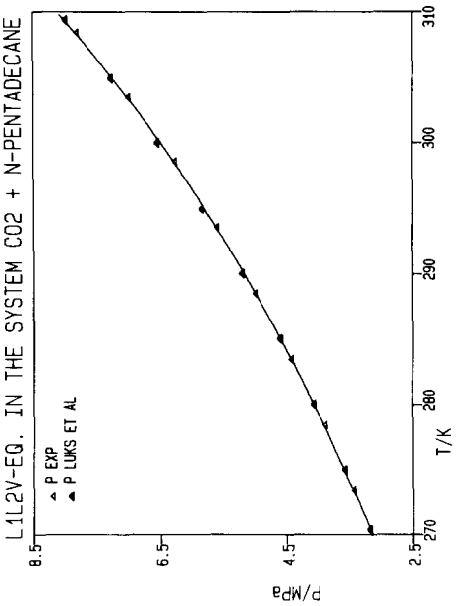
Calculated 95%-confidence intervals and average absolute deviations (AAD) of the data obtained for the L_1L_2V equilibria of [(1-x) $C_{14}H_{30}$ + x CO_2].

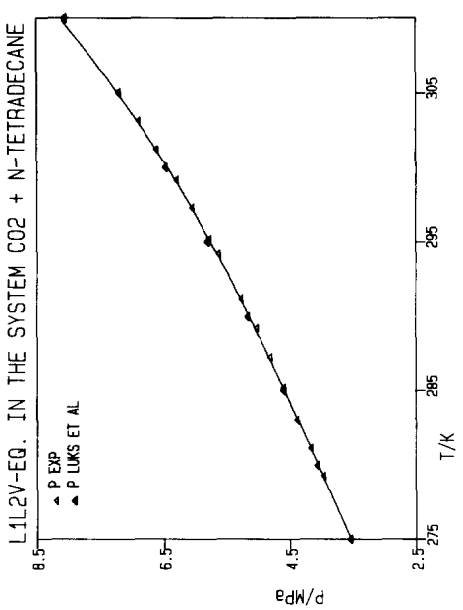
T/K	Δx_V	Δx_{L_2}	Δx_{L_1}	$\Delta\rho_V/gml^{-1}$	$\Delta\rho_{L_2}/gml^{-1}$	$\Delta\rho_{L_1}/gml^{-1}$
279.26	0.0048	0.0072	0.0015	0.0125	0.0175	0.0249
281.14	0.0061	0.0050	0.0020	0.0107	0.0151	0.0218
283.08	0.0070	0.0057	0.0024	0.0135	0.0177	0.0271
285.16	0.0081	0.0066	0.0031	0.0172	0.0213	0.0344
287.24	0.0030	0.0046	0.0015	0.0120	0.0140	0.0249
289.21	0.0057	0.0021	0.0018	0.0100	0.0078	0.0156
291.18	0.0114	0.0029	0.0035	0.0133	0.0113	0.0194
294.21	0.0107	0.0026	0.0031	0.0170	0.0108	0.0215
295.17	0.0038	0.0010	0.0013	0.0060	0.0040	0.0083
297.26	0.0032	0.0009	0.0015	0.0054	0.0036	0.0092
299.19	0.0050	0.0012	0.0019	0.0083	0.0050	0.0108
301.15	0.0018	0.0006	0.0008	0.0043	0.0062	0.0058
303.13	0.0037	0.0028	0.0022	0.0147	0.0102	0.0233
AAD	0.0057	0.0033	0.0020	0.0111	0.0111	0.0190

compositions and densities to relative great changes in the parameters, greater than the accuracy with which these parameters were obtained, was shown to be of minor importance. The introduction of an error of 0.03 g in the amounts of the components, 0.15 mm in the cathetometer readings and 1 % in the calibration constants of the Cailletet tube, errors which are about three times the actual experimental errors, caused changes in the calculated mole fractions and densities within the accuracy stated above. It should be noted that the minimum amount of a component used in the experiments was 2 g and that the minimum height difference between the fluid phases was 1 cm, corresponding to a minimum phase volume of 3 ml. The variance in the temperature between the various experiments is less then 0.1 K. From Fig. 3 it follows that $\delta x_{i,j}/\delta T = 0.0079\ K^{-1}$. The variance in composition due to the variance in temperature is therefore less then 0.0008. This is also negligible.

The barotropic inversion point on the three phase locus of all systems is obtained from Fig. 4. The barotropic inversion points are:

CO_2 + tetradecane: 293.2±0.2 K
CO_2 + pentadecane: 291.7±1.0 K
CO_2 + hexadecane : 290.8±0.1 K





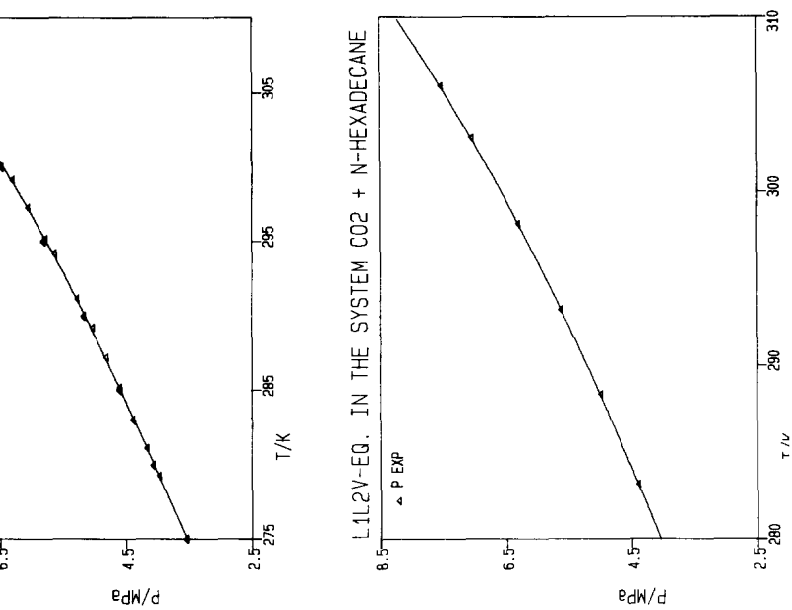


Fig. 5. Experimental and predicted pressures (see text) of the system [(1-x) C_nH_{2n+2} + x CO_2] for n=14, 15, 16 at three phase equilibrium L_1L_2V as a function of temperature. Δ, this work; Δ, Hottovy et al. (1981); ----, Peng-Robinson equation of state. a: CO_2 + tetradecane; b : CO_2 + pentadecane; c: CO_2 + hexadecane.

TABLE 5

Calculated 95%-confidence intervals and average absolute deviations (AAD) of the data obtained for the L_1L_2V equilibria of [(1-x) $C_{15}H_{32}$ + x CO_2].

T/K	Δx_V	Δx_{L_2}	Δx_{L_1}	$\Delta\rho_V$/mlg^{-1}	$\Delta\rho_{L_2}$/mlg^{-1}	$\Delta\rho_{L_1}$/mlg^{-1}
273.42	0.0203	0.0097	0.0019	0.0181	0.0232	0.0148
278.48	0.0170	0.0085	0.0021	0.0163	0.0228	0.0149
283.47	0.0176	0.0081	0.0025	0.0188	0.0242	0.0169
288.49	0.0130	0.0068	0.0021	0.0185	0.0205	0.0157
293.54	0.0062	0.0075	0.0030	0.0136	0.0219	0.0266
298.55	0.0022	0.0040	0.0005	0.0098	0.0098	0.0081
303.48	0.0016	0.0043	0.0006	0.0099	0.0103	0.0092
308.43	0.0019	0.0052	0.0012	0.0121	0.0132	0.0121
AAD	0.0007	0.0012	0.0008	0.0116	0.0023	0.0102

TABLE 6

Calculated 95%-confidence intervals and average absolute deviations (AAD) of the data obtained for the L_1L_2V equilibria of [(1-x) $C_{16}H_{34}$+ x CO_2].

T/K	Δx_V	Δx_{L_2}	Δx_{L_1}	$\Delta\rho_V$/mlg^{-1}	$\Delta\rho_{L_2}$/mlg^{-1}	$\Delta\rho_{L_1}$/ml^{-1}
283.17	0.0174	0.0112	0.0034	0.0220	0.0269	0.0308
288.33	0.0179	0.0176	0.0035	0.0387	0.0390	0.0411
293.18	0.0104	0.0114	0.0023	0.0266	0.0267	0.0276
298.13	0.0047	0.0089	0.0019	0.0167	0.0205	0.0232
303.11	0.0033	0.0106	0.0015	0.0204	0.0223	0.0232
306.10	0.0033	0.0071	0.0022	0.0157	0.0172	0.0191
AAD	0.0028	0.0020	0.0003	0.0079	0.0012	0.0066

TABLE 7

L_1L_2V equilibria in [(1-x) C_nH_{2n+2} + x CO_2]. Calculated optimal binary interaction parameters and variances of the fit to the Peng-Robinson (PR) and the modified Peng-Robinson (PRmod) equations of state.

systeem	eos	k_{12}	l_{12}	var p/Mpa	var x_{L_2}	var x_{L_1}
CO_2+C_{14}	PR	0.09	0.0090	0.015	0.0169	0.0134
CO_2+C_{15}	PR	0.10	0.0087	0.050	0.0178	0.0086
CO_2+C_{16}	PR	0.10*	0.0095	0.030	0.0042	0.0032
CO_2+C_{16}	PR	0.10*	0.0000	0.024	0.0136	0.0026
CO_2+C_{14}	PRmod	0.09	0.0130	0.024	0.0192	0.0150
CO_2+C_{15}	PRmod	0.10	0.0011	0.055	0.0191	0.0094
CO_2+C_{16}	PRmod	0.10*	0.0012	0.028	0.0057	0.0039
CO_2+C_{16}	PRmod	0.10*	0.0000	0.016	0.0225	0.0047

*Gardner et al., 1981

The obtained compositional data have been fitted to the well-known Peng-Robinson equation of state (eqns. (4) to (8)) (Peng and Robinson, 1976) and a modified Peng-Robinson equation of state (Tsonopoulos and Heidman, 1986) using the classical van der Waals mixing rules.

$$p=RT/(V-b)-a(T)/[V*(V+b)+b*(V-b)] \quad (4)$$

with

$$a(T)=0.45724*R^2*T_c^2/p_c*\alpha \quad (5)$$

and

$$b=0.0778*R*T_c*/p_c \quad (6)$$

The temperature dependance of α follows from eqn. (7)

$$\alpha=[1+m*(1-T_r)^{0.5}]^2 \quad (7)$$

with

$$m=0.37464+1.54226*\omega-0.26992*\omega^2 \quad (8)$$

The original Peng-Robinson equation of state uses eqn. (8) in which m is a quadratic function of the acentric factor ω. This function has been derived by fitting vapor-pressure data of substances with $\omega \leq 0.5$. Later Robinson and Peng (1978) modified this equation for substances with $\omega \leq 1$. This resulted in eqn. (9):

$$m=0.379642+1.48503*\omega-0.164423*\omega^2+0.016666*\omega^3 \quad (9)$$

The following mixing rules were applied

$$a=a_1x_1^2+2x_1x_2(a_1a_2)^{1/2}(1-k_{12})+a_2x_2^2 \quad (10)$$

$$b=b_1x_1^2+2x_1x_2(b_1+b_2)(1-1_{12})+b_2x_2^2 \quad (11)$$

For l_{12}=0 eqn. (11) reduces to the more conventional mixing rule $b=x_1b_1+x_2b_2$.

The equations of state where compared by estimating the variances:

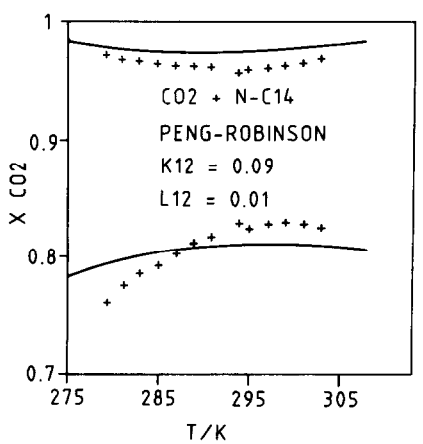
$$\text{var } p = 1/(n_d-n_p) * \sum_{i=1}^{n} (p_{cal}-p_{exp})^2 \qquad (12)$$

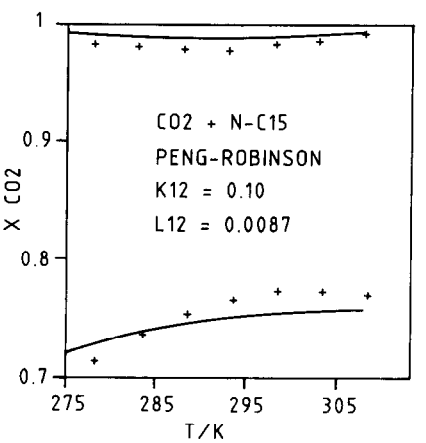
$$\text{var } x_i = 1/(n_d-n_p) * \sum_{i=1}^{n} (x_{cal}-x_{exp})^2 \qquad (13)$$

with: n_d number of data points

n_p number of binary interaction parameters

Using a parameter $l_{12}\neq 0$ results in a much better description of experimental L_1L_2V equilibria, as is illustrated in Table 7 for the systen CO_2 + hexadecane.





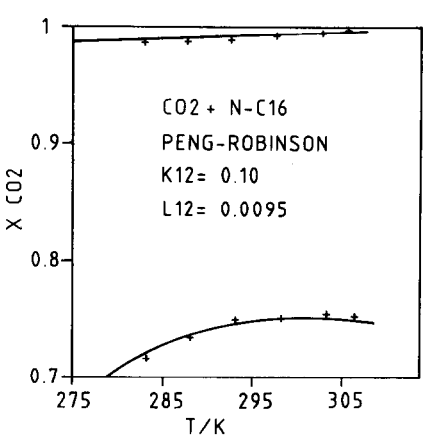


Fig. 6. Predicted compositions of the liquid phases of the systems [(1-x) C_nH_{2n+2} + x CO_2] for n=14, 15, 16 at three phase equilibria L_1L_2V as a function of temperature. +, experiments; ----, Peng-Robinson equation of state. a: CO_2 + tetradecane; b: CO_2 + pentadecane; c: CO_2 + hexadecane.

From Table 7 it follows that there is no significant difference in the quality of the fit of the experimental data using the original and modified Peng-Robinson equations of state. The calculated mole fractions CO_2 are plotted in Fig. 6 and the calculated pressures in Fig. 5. The calculated liquid phase densities differ about 15% from the experimental values, but the barotropic inversion points are predicted accurately. With the obtained parameters for the Peng-Robinson equation the liquid-liquid and liquid-vapor equilibria of the system CO_2 + hexadecane were calculated and compared with the experimental data published by Charoemsumbut-Amon et al. (1986) and Sebastian et al. (1980). The L_1-L_2 and L-G equilibria of the system CO_2 + hexadecane calculated with the Peng-Robinson equation of state using the parameters k_{12} an l_{12} from Table 7 did not differ more than 0.01 in the mole fraction CO_2 up to pressures of 100 bar and temperatures up to 350 ^oC.

CONCLUSIONS

The volumetric method as proposed by Knobler and Scott (1980) has proven to be applicable to the determination of fluid phase equilibria in CO_2 + n-alkane systems. It is shown in this paper which errors can occur and also how these errors can be prevented or corrected. The modified Peng-Robinson equation of state does not predict the phase behaviour of the CO_2 + n-alkane systems investigated significantly better than the original Peng-Robinson equation of state.

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